

Advanced Generic Definitions

[BGA]

Example 1 : Generic Three - Way Product

Generic function for three-component tuples:

$[X, Y, Z]$	$\frac{}{triple : X \times Y \times Z \rightarrow X \times Y \times Z}$ $\frac{}{getFirst : X \times Y \times Z \rightarrow X}$ $\frac{}{getSecond : X \times Y \times Z \rightarrow Y}$ $\frac{}{getThird : X \times Y \times Z \rightarrow Z}$
	$\forall x : X; y : Y; z : Z \bullet$ $triple(x, y, z) = (x, y, z) \wedge$ $getFirst(x, y, z) = x \wedge$ $getSecond(x, y, z) = y \wedge$ $getThird(x, y, z) = z$

Projection functions for triples.

Example 2 : Generic Binary Tree (Conceptual)

Binary trees are typically defined for specific types in Z notation rather than as fully polymorphic structures. For a concrete type like N, you would define: $Tree ::= leaf \mid node(N \text{ cross } Tree \text{ cross } Tree)$. Generic trees require complex workarounds beyond standard gendef blocks, as free types in fuzz are not easily parameterized.

Example 6 : Generic State Schema

A generic container with capacity: Note: Fuzz supports generic schemas using the $schema[X]$ syntax (not gendef):

$Container[X]$	$\frac{}{contents : seq\ X}$ $\frac{}{capacity : \mathbb{N}}$ $\frac{}{\#contents \leq capacity}$
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This schema is parameterized by the element type X. It can be instantiated as $Container[N]$, $Container[Char]$, etc.

Example 7 : Generic Relation Operations

Relational composition for any types:

$[X, Y, Z]$	$\frac{}{compose : (X \leftrightarrow Y) \times (Y \leftrightarrow Z) \rightarrow (X \leftrightarrow Z)}$ $\frac{}{\forall R : X \leftrightarrow Y; S : Y \leftrightarrow Z \bullet}$ $\frac{}{compose(R, S) = R \circ S}$
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Composes two relations using forward composition.

Example 8 : Generic Option Type (Conceptual)

Optional values (like Maybe or Option in other languages) are challenging to express generically in Z notation due to limitations with parameterized free types. For a specific type, you would define: $\text{Option} ::= \text{none} \mid \text{some}(T)$. Fully generic option types with operations require advanced patterns beyond standard gendef blocks.

Example 9 : Generic Stack Schema

Note: Define generic schema separately, then declare operations in gendef:

$\frac{\text{Stack}[X]}{\text{items} : \text{seq } X}$
$\text{maxSize} : \mathbb{N}$
$\# \text{items} \leq \text{maxSize}$
$\frac{[X]}{\text{emptyStack} : \text{Stack}[X]}$
$\text{stackSize} : \text{Stack}[X] \rightarrow \mathbb{N}$
$\text{emptyStack.items} = \langle \rangle \wedge \text{emptyStack.maxSize} = 100$
$\forall s : \text{Stack}[X] \bullet \text{stackSize}(s) = \#s.\text{items}$

Generic stack with query operations. The schema is defined separately using schema[X] syntax, then operations use gendef.

Example 10 : Generic Pair Schema

Use schema[X, Y] syntax for generic schemas:

$\frac{\text{OrderedPair}[X, Y]}{\text{first} : X}$
$\text{second} : Y$

A generic pair type. The schema has two type parameters and can be instantiated as OrderedPair[N, Char], etc.

Example 11 : Generic Zip Function

Combine two sequences into a sequence of pairs:

$\frac{[X, Y]}{\text{zip} : \text{seq } X \times \text{seq } Y \rightarrow \text{seq } (X \times Y)}$
$\text{zip}(\langle \rangle, \langle \rangle) = \langle \rangle$
$\forall x : X; y : Y; xs : \text{seq } X; ys : \text{seq } Y \bullet$
$\text{zip}(\langle x \rangle \hat{\ } xs, \langle y \rangle \hat{\ } ys) =$
$\langle (x, y) \rangle \hat{\ } \text{zip}(xs, ys)$
$\forall xs : \text{seq } X \bullet \text{zip}(xs, \langle \rangle) = \langle \rangle$
$\forall ys : \text{seq } Y \bullet \text{zip}(\langle \rangle, ys) = \langle \rangle$

Pairs up corresponding elements from two sequences.

Example 12 : Practical Example - Generic Database TableGA

Note: Define generic schema separately, then query operations in gendef:

$[Key, Value]$

$TableGA[Key, Value]$ $entries : Key \rightarrow Value$ $size : \mathbb{N}$
$size = \#(\text{dom } entries)$

$[Key, Value]$ $emptyTable : TableGA[Key, Value]$ $tableSize : TableGA[Key, Value] \rightarrow \mathbb{N}$ $allKeys : TableGA[Key, Value] \rightarrow \mathbb{P} Key$
$emptyTable.entries = \{\} \wedge emptyTable.size = 0$ $\forall t : TableGA[Key, Value] \bullet tableSize(t) = t.size$ $\forall t : TableGA[Key, Value] \bullet allKeys(t) = \text{dom } t.entries$

A generic key-value table with query operations. This demonstrates how generic definitions enable reusable, type-safe data structures. The TableGA schema is defined with schema[Key, Value] syntax, allowing proper fuzz typechecking.